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## **AU/ME-7001-CBGS**

### **B.E. VII Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **Mechanical Vibrations**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is the vibration? Explain the Rayleigh's method. What are its application?  
b) A force  $F_0 \sin \omega t$  acts on displacement  $X_0 \sin(\omega t - 60^\circ)$ , where  $F_0 = 100 \text{ N}$   $X_0 = 0.02 \text{ m}$ ,  $\omega = 2\pi \text{ rad/s}$ . Find the work done during
  - i) First cycle
  - ii) First second
  - iii) First  $1/40$  second
  - iv) First quarter second
2. a) What is resonance? A body performs two motions simultaneous  $x_1 = 1.9 \sin(9.5 t)$   $x_2 = 2.1 \sin(10.0 t)$  Find the maximum and minimum amplitude of combined motion and time period of periodic motion in cm and second.  
b) Determine the natural frequency of system shown in figure by Newton's method. Assume  $M$  is the mass of disk and  $J_0$  is polar moment of inertia of disk about centre. System is in static equilibrium.

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3. Show that the maximum velocity of vibration of a mass of a spring-mass-dashpot system occurs at  $(\omega/\omega_n) = 1$  irrespective of the amount of damping.
4. Determine the critical speed of a 1000 kg automobile travelling on a concrete road with expansion joints spaced 12 metres apart, if the static deflection of the spring system is 50mm.
5. Deduce the equation of a whirling of light flexible shaft with an unbalanced disk at the centre of its length with a without damping.
6.
  - a) Distinguish between longitudinal, transverse and torsional vibration. Design beat phenomena.
  - b) A vibratory system is defined by following parameters  
 $m = 3\text{kg}$ ,  $k = 100\text{ N/m}$ ,  $c = 3\text{ N- sec/m}$   
Calculate
    - i) Damping factor
    - ii) Damped natural frequency
    - iii) Logarithmic decrement
    - iv) Ratio of two consecutive amplitudes
7.
  - a) Distinguish between viscous damping and Coulomb damping along with examples. Also explain vibration isolation.
  - b) Discuss free vibration and forced vibrations of undamped system.
8. Write short notes on the following:
  - a) Principle of velocity and acceleration measuring instruments
  - b) Frequency response plots
  - c) Phase shift plots

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